Project Plan: University Teaching Time Optimization

Scenario 2: TIME

Group Number: 17

Team Members: Yang Feng (s2775736), Zongyuan Wang (s2794231),

Yumeng Yao (s2792276), Yifei Wang (s2328864)

1. Project Overview

Background:

The University currently operates under a standard teaching times policy, with core teaching hours spanning Monday to Friday across various timeslots. However, there is growing interest in exploring whether a more compressed teaching schedule could improve student experience while maintaining operational efficiency. This project investigates the feasibility of reducing core teaching hours, either by limiting daily teaching to 9am-5pm or by eliminating Friday afternoon sessions, while ensuring educational quality, minimizing scheduling conflicts, and providing adequate break times for students.

Research Questions:

Q1: Would it be possible to reduce core teaching hours to Mon-Fri 9am-5pm?

Q2: Would it be possible to eliminate teaching on Friday 12pm-6pm (Mon-Thu remain the same)?

Q3: Could core teaching/lectures still be delivered without clashes in each scenario?

Q4: Could we provide a lunch break for majority of students between 12pm-2pm?

Q5: How would these changes affect timeslot/room utilization throughout the week?

Success Criteria:

The project will be considered successful if we can:

(a) quantify the impact on room and timeslot utilization rates under each proposed time constraint scenario, comparing against current baseline metrics;

(b) identify and measure the number and severity of teaching clashes that would arise under reduced hours, particularly for core lectures and compulsory modules;

(c) determine the feasibility of providing a lunch break between 12pm and 2pm for the majority of students, analyzing the proportion of students who could benefit;

(d) provide actionable recommendations supported by clear data visualizations including utilization heatmaps, clash frequency charts, and comparative scenario analyses.

Technical Approach:

We will use **Python** (pandas, numpy) for data extraction, cleaning, and exploratory analysis, and **FICO Xpress Python interface** for building optimization models to evaluate timetable feasibility under various time constraints. Visualization will be done using matplotlib and seaborn.

The optimisation will follow a feasibility-first approach, prioritising the satisfaction of hard constraints before evaluating soft objectives such as lunch breaks and utilisation balance.
Lunch break availability will be treated as a soft constraint and evaluated through post-optimisation metrics rather than enforced as a hard constraint, to avoid infeasibility.

2. Task List

The project is divided into 4 phases with 12 tasks. Each task is assigned to specific team members with clear deliverables and dependencies. The table below summarizes responsibilities and maps each task to the research questions addressed.

ID	Task	Resp	Data Files	Key Fields	Deliverable	Deps	Q
Phase 1: Data Preparation (W1-W2)							

T1	Data collection & cleaning	YF	All Excel files	All - validate, clean, join	Clean dataset	-	**-**
T2	Baseline utilization analysis	ZW	Event Module Room, Rooms	Timeslot, Duration, Room, Capacity	Baseline report	T1	**Q5**
T3	Literature review	YY	-	-	Literature summary	-	**-**
T4	Constraint framework design	YW	Event Module Room	Event Type, Timeslot, Duration	Constraint spec	T3	**-**
Phase 2: Time Constraint Analysis (W3-W5)							
T5	9am-5pm feasibility analysis	YF	Event Module Room, Student Event	Timeslot (filter 9-17), Event ID	9-5 impact report	T1-T4	**Q1**
T6	No-Friday-afternoon analysis	ZW	Event Module Room, Student Event	Timeslot (excl Fri 12-18)	Fri-PM report	T1-T4	**Q2**
T7	Clash detection & analysis	YY	Student Event, Event Module Room	AnonID, Event ID, Timeslot	Clash report	T5,T6	**Q3**
T8	Lunch break feasibility	YW	Student Event, Event Module Room	AnonID, Timeslot (12-14 gap)	Lunch report	T5,T6	**Q4**
Phase 3: Analysis & Visualization (W5-W7)							
T9	Utilization impact comparison	YF, ZW	Event Module Room, Rooms	Room, Timeslot, Capacity	Utilization report	T5-T8	**Q5**
T10	Data visualization & dashboard	YY, YW	All analysis outputs	Aggregated metrics	Visualizations	T7-T9	**Q1-5**
T11	Sensitivity analysis	ZW, YW	Event Module Room, Student Event	Time-related scheduling information; student–event mappings	Sensitivity report	T5–T8	**Q1–4**
Phase 4: Integration & Reporting (W7-W9)							
T12	Integration & validation	All	All outputs	Cross-validation	Validated results	T10	**All**
T13	Final report writing	All	All deliverables	-	Final report	T12	**All**

**Note: Resp = Responsibilities, Deps = Dependencies, Q = Research Question addressed.*

3. Project Schedule

The project spans about 9 weeks from February 10 to April 10, 2026. The Gantt chart below shows task timelines and four key milestones marking phase completions.

Task	**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8**	**W9**
	2/10	2/17	2/24	3/3	3/10	3/17	3/24	3/31	4/7
T1: Data collection									
T2: Baseline analysis									
T3: Literature review									
T4: Constraint framework									
T5: 9am-5pm analysis									

T6: No-Fri-afternoon			■	■					
T7: Clash detection				■	■				
T8: Lunch break				■	■				
T9: Utilization					■	■			
T10: Visualization						■	■		
T11: Sensitivity analysis						■	■		
T12: Integration							■	■	
T13: Final report								■	■
Milestones		◆			◆		◆		◆

Milestones:

M1 (Feb 21): Data collection & baseline analysis ready

M2 (Mar 14): Modelling & Analysis complete

M3 (Mar 28): Visualization ready

M4 (Apr 10): Final report complete

4. Risk Register

Five key risks have been identified in our project, with mitigation strategies and contingency plans for each. The highest priority risk (R3) addresses the possibility that reduced teaching hours may be infeasible.

ID	Risk	P	I	Mitigation	Contingency
R1	Data quality: Missing/inconsistent records	M	H	Early validation; document assumptions	Use clean subset
R2	Computational complexity: Slow on large data	M	M	Sample first; optimize queries	Report approximations
R3	Infeasibility: Cannot fit teaching in reduced hours	H	H	Early check; identify critical events	Quantify minimum hours
R4	Team unavailability	L	M	Cross-training; documentation	Redistribute tasks
R5	Scope creep: Additional questions	M	M	Clear scope; change process	Defer to future
R6	Interpretation risk: Results may be sensitive to modelling assumptions (e.g. definition of clashes or lunch breaks)	M	M	Explicitly state assumptions; compare multiple definitions	Report results under alternative assumptions

**Note: P = Probability, I = Impact. Rating: L = Low, M = Medium, H = High. Score = P × I. Priority: R3(9), R1(6), R2(4), R5(4), R4(2).*

5. Critical Path Conclusion

The total project duration is 9 weeks (equivalent to 18 half-week time units).

The critical path (i.e., activities with zero float / zero slack) is:
T1&T3(start)→T2 &T4 → (T5 and T6 in parallel) → (T7 and T8 in parallel) → T9 → T10(T11) → T12